1. Assessment of Diffusion-Generated Data for Retraining

After evaluating the proposal to relabel the million-image diffusion dataset, I recommend against using this data for model retraining. The core issue is that diffusion models introduce structural distortions—such as dissolved object boundaries and hallucinated geometries—that are not present in standard sensor corruptions like noise or motion blur.

Because these distortions fundamentally alter object presence, our existing ground truth is no longer valid. Proceeding with a relabeling effort of this scale shifts our focus from "fault-injection testing" to "large-scale synthetic dataset construction," which is outside the current project scope. Furthermore, training on these unrealistic geometric patterns may negatively bias the warehouse vehicle's perception system (intended deployment) against real-world sensor data.

2. Proposed Robustness Evaluation Workflow

To ensure our TensorRT engine remains aligned with actual deployment behavior, we will use the diffusion dataset strictly as a stress-test for the existing system.

Phase I: Inference and Resource Logging

We will process each fault type independently as still frames to isolate results and avoid video encoding artifacts. During execution on the deployed engine, we will capture the following:

- **Performance Metrics:** Frame-level latency, detection counts post-processing, and flags for degraded outputs.
- **System Health:** GPU utilization, RAM usage, power consumption, and thermal data via jtop.

Phase II: Offline Analysis and Ranking

Following inference, we will analyze the CSV logs to establish a baseline for system stability:

- **Stability Analysis:** Calculation of p95/p99 tail latency and jitter.
- **Failure Rates:** Measuring temporal detection stability (flicker) and the invalid frame rate.
- **Severity Ranking:** Fault types will be ranked by their impact on resource stress and detection stability.

3. Recommendations and Guardrails

The results of this evaluation should be used to develop runtime safeguards. This includes establishing thresholds for confidence gating or fallback mechanisms when the system encounters extreme instability.

f accuracy metrics (mAP, precision, recall) are still required, we should generate a separate dataset using physically plausible transformations that preserve original object geometry. This allows for a meaningful benchmark without the need for expensive relabeling.

Conclusion

The diffusion dataset is a high-value tool for identifying failure modes but is unsuitable for retraining. This workflow provides measurable insights into system robustness while maintaining our current deployment goals.